

SYSTEM OVERVIEW

The system will take the Iris dataset (150 flowers with 4 measurements each), use clustering algorithms to automatically group the flowers into categories, and then compare the computer's groupings with the actual species labels to see how accurate the clustering was. No GUI needed. This is a data analysis program that runs and shows results in the console or terminal.

PART 1: DATA PREPROCESSING

What your lecturer wants is to standardize numerical features to eliminate dimension influence. Different measurements have different scales. Sepal length might be 5.1 cm while petal width might be 0.2 cm. Without standardization, sepal length would unfairly influence the clustering because it has larger numbers. You need to load the Iris dataset which has 150 flowers and 4 measurements each. Then extract the 4 features which are sepal length, sepal width, petal length, and petal width. After that, apply standardization to transform all values to the same scale so the result should be values centered around 0 with standard deviation of 1. You must load the dataset correctly and apply standardization before any clustering. For error handling, if the dataset fails to load you should display an error message and exit. If the data contains null or empty values you should display an error message and exit.

PART 2: K-MEANS CLUSTERING

What your lecturer wants is to implement K-Means clustering. K-Means is an algorithm that groups similar data points together. You tell it that you want 3 groups and it finds 3 natural clusters in the data. The algorithm randomly picks 3 points as centers. Every flower is assigned to the nearest center. The centers move to the middle of their assigned flowers. These steps repeat until groups stop changing. You need to set the number of clusters to 3 because the Iris dataset has 3 real species. Set a random state to a fixed number like 42 so results are repeatable. Run the clustering on the standardized data from Part 1. You must use 3 clusters and run K-Means on standardized data. It is not required but good to print the cluster centers and show how many flowers are in each cluster. For error handling, if K-Means fails to converge you should display an error message and exit.

PART 3: HIERARCHICAL CLUSTERING

What your lecturer wants is to implement hierarchical clustering. Hierarchical clustering builds a tree called a dendrogram showing how flowers are related. It starts with each flower as its own group and repeatedly merges the two most similar groups. You need to use the Ward linkage method which minimizes variance within clusters. Use Euclidean distance metric which is the straight-line distance between points. Run this on the standardized data from Part 1. You must generate a dendrogram tree diagram and use it to show how clusters form. It is optional but helpful to draw a horizontal line on the dendrogram at the height that gives 3 clusters. For error handling, if linkage calculation fails you should display an error message and exit.

PART 4: MODEL EVALUATION

What your lecturer wants is to use silhouette coefficient and Rand index to evaluate clustering quality. The silhouette coefficient measures how well-separated and tight the clusters are. The score ranges from -1 to +1 and a score close to +1 means well-separated clusters. The Rand Index measures how similar the computer clusters are to the real species labels. The score ranges from 0 to 1 and a score close to 1 means it matches the real species. You need to calculate the silhouette coefficient for the K-Means results. Then calculate the Rand Index comparing K-Means clusters to the real species. Print both scores clearly. You must calculate both metrics and print or display the results. A silhouette score near 0.5 to 0.7 indicates good clustering. A Rand index near 0.7 to 0.9 indicates good match with real species. You should explain what the numbers mean in plain English. For error handling, if real labels are missing you cannot calculate Rand Index so display an error message.

PART 5: RESULT VISUALIZATION

What your lecturer wants is to visualize clustering results and analyze feature distribution. Create pictures or graphs that show what the computer found so you can see if the clusters make sense. You must create a scatter plot of K-Means clusters showing flowers in 3 different colors based on computer groups. Use the first two features which are sepal length and sepal width. Color each point by its assigned cluster using 3 different colors. Add a title and axis labels. You must also create a dendrogram tree diagram from hierarchical clustering showing the full tree structure with a title. It is optional to create a pairplot of features or a comparison plot showing computer clusters versus real species side by side. You must create at least one scatter plot showing K-Means clusters and create a dendrogram from hierarchical clustering. For error handling, if plotting fails because there is no display available, save the plots as image files instead.

COMPLETE SYSTEM REQUIREMENTS SUMMARY

What your system **MUST** do includes loading the Iris dataset, standardizing the features, running K-Means with 3 clusters, running Hierarchical clustering, calculating the Silhouette Coefficient, calculating the Rand Index, creating a scatter plot of K-Means clusters, creating a dendrogram, and printing or displaying the evaluation results. What your system does **NOT** need includes a GUI interface, user input fields, Groups A-E, 0 to 40 range validation, a web interface, database storage, or real-time classification.

DATA VALIDATION AND ERROR HANDLING

You need to implement several error checks. If the dataset file does not exist, display an error message saying "ERROR: Could not load Iris dataset" and terminate. If the dataset does not have 150 rows, display "ERROR: Dataset has incorrect number of samples" and terminate. If the dataset does not have 4 features, display "ERROR: Dataset has incorrect number of features" and terminate. If there are null or missing values, display "ERROR: Dataset contains missing values" and terminate. If K-Means fails to converge, display "ERROR: K-Means failed to converge" and terminate. If hierarchical linkage fails, display "ERROR: Hierarchical clustering failed" and terminate. For a successful run, the output message should show that all tasks are completed, the dataset loaded with 150 samples and 4 features, standardization complete, K-Means clustering complete with 3 clusters, hierarchical clustering complete, the silhouette coefficient score, the Rand index score, and that visualizations have been saved.

SAMPLE OUTPUT EXPECTATION

When you run your program, the console should show something like this. First a title line saying IRIS FLOWER CLUSTERING ANALYSIS. Then loading the Iris dataset showing that 150 samples with 4 features were loaded and real species labels are Setosa, Versicolor, and Virginica. Then data preprocessing showing that standardization is complete. Then K-Means clustering showing that it is complete and cluster sizes such as 50, 50, and 50. Then hierarchical clustering showing that the dendrogram is generated. Then model evaluation showing the silhouette coefficient score and Rand index score. Then visualizations showing that the images have been saved. Finally an analysis section explaining that a silhouette score of 0.55 indicates moderately well-separated clusters, a Rand index of 0.73 shows 73 percent agreement with real species labels, and that clustering successfully distinguished Setosa but had some confusion between Versicolor and Virginica.

IMPLEMENTATION ORDER

Here is the order of what to code first to last. First load the Iris dataset and print confirmation. Second check the data for errors such as nulls or wrong size. Third standardize the features. Fourth run K-Means clustering. Fifth print K-Means results including cluster sizes. Sixth run Hierarchical clustering and generate the dendrogram. Seventh calculate the Silhouette Coefficient. Eighth calculate the Rand Index. Ninth create a scatter plot of K-Means clusters. Tenth print all results with analysis.

FINAL CHECKLIST BEFORE SUBMITTING

Check that you can load the Iris dataset without errors. Check that you standardized the data before clustering. Check that you ran K-Means with 3 clusters. Check that you ran Hierarchical clustering. Check that you calculated the Silhouette Coefficient. Check that you calculated the Rand Index. Check that you created a scatter plot of clusters. Check that you created a dendrogram. Check that your code has error handling. Check that your output explains what the scores mean.